

VIBHAV INNA KEDEGE

MSc Embedded Systems (Honours), BSc Electrical Engineering

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📍 Rijswijk, The Netherlands

• Nationality: Dutch (Netherlands, EU) & Indian (OCI)



PROFESSIONAL EXPERIENCE

Alfen N.V.

DevOps Software Engineer | R&D Department

📅 Oct 2023-Present 📍 Almere, The Netherlands

- Engineering distributed test automation platforms using **Python, Jenkins, Docker** on **Linux** systems that is accelerating firmware delivery by **6x** (from 12 weeks to 2 weeks).
- Architecting real-time analytics system (**FastAPI, PostgreSQL**) to automate analysis of **5000+** daily tests, enable defect detection **10 weeks earlier** and recover **EUR 150K+** of costs/quarterly.
- Building **MLOps** predictive maintenance system for EV charger infrastructure, enabling proactive defect detection to reduce field maintenance costs.

KPN (via Capgemini)

Data Engineer (Big Data) | KPN In-Home Analytics

📅 Oct 2021-Sep 2023 📍 The Hague, The Netherlands

- Transformed raw internet quality logs into actionable business intelligence (**Tableau and Python**), enabling data-driven operational decisions for executive leadership.
- Optimized scalable data pipelines (**Spark and Scala**) processing **1M+ IoT devices**, enabling reduced network downtime through proactive infrastructure monitoring.
- Developed real-time ML system (**Spark MLlib**) to detect service degradation and reduce customer churn through proactive intervention.

Almende B.V.

Robotics R&D Engineer | R&D Department

📅 Jun 2020-Aug 2021 📍 Rotterdam, The Netherlands

- Architected distributed multi-agent algorithms (**Python, C++, and ROS**) for autonomous search and rescue robots in unknown environments, validating theoretical coordinated behavior through simulation before field deployment (Link)
- Published peer-reviewed research (ICAART 2022) on **Multi-Agent Path Planning** that aimed to bridge the gap from simulation to field deployment.(Link)

Alstom Transport

System Engineer - Train Networks | Train Control Department

📅 Jul 2017-Jul 2019 📍 Bangalore, India and Sydney, Australia

- Validated Ethernet-based **Autonomous Train Control Systems** across India and Australia; Achieved a 90% validation test pass rate 3 months before schedule, exceeding customer expectations and reducing project risk.

PERSONAL STATEMENT

A proactive Software Engineer with 7+ years of experience specializing in **Data Platforms and MLOps**. Proven track record of designing distributed systems for **1M+ devices** and accelerating software delivery pipelines by **6x** through scalable infrastructure engineering. '

EDUCATION

Delft University of Technology

MSc Embedded Systems (Honours)

📅 2019 - 2021 📍 Delft, Netherlands

- Specialisation - Artificial Intelligence (AI) and Robotics.
- Thesis - "Multi Robot Exploration and Planning in limited communication environments" (Link).
- Published at ICAART 2022 (Link).

National Institute of Technology

Karnataka (NITK-Surathkal)

Bachelor of Technology, Electrical and Electronics Engineering

📅 2013 - 2017 📍 Mangalore, India

- Specialisation - Embedded Software
- Thesis - "Multiple Interface Automation using Distributed Embedded Devices" (Link)

TECHNICAL SKILLS

Core Languages

Python SQL C++ Scala Java

Data & AI / MLOps

Apache Spark Tensorflow MLflow

Prefect MinIO LLMs RAG

Platform & Cloud

Docker Jenkins Azure FastAPI

Linux

Engineering & Systems

Distributed Systems System Design

HIL Testing CI/CD

SELECTED PROJECTS

ML-Powered Stock Prediction & Monitoring System

MLOps Reference Architecture

📅 Jan 2026 – Present

- Engineering a production **MLOps** platform processing S&P500 stock market data to establish reusable patterns for production ML systems.
- Building end-to-end ML infrastructure using **Python, Docker, and PostgreSQL** with experiment tracking (**MLflow**), workflow orchestration (**Prefect**), prediction serving (**FastAPI**), and real-time monitoring dashboard (**Streamlit**) ([Live Demo](#))

Local RAG & LLM Orchestration Engine

Technical Case Study

📅 Jul 2025

- Built privacy first system (**Ollama, LangChain, RAG**) eliminating cloud dependencies and data exposure.
- Designed context-injection framework reducing LLM hallucinations through semantic retrieval for reliable local AI. ([Technical Article](#))

Vision-Based Logistics Automation Pipeline

Freelance MLOps Project

📅 Apr 2023 - Jul 2023

📍 Delft, Netherlands

- Designed ML workflow (**YOLOv8, SAM**) for **100K+** images, reducing iteration cycles from weeks to daily via CI/CD..
- Optimized inference pipeline reducing computational costs and enabling production-scale logistics automation.

Reinforcement Learning Hyperparameter Optimization

Technical Case Study

📅 Sep 2020

📍 Delft, Netherlands

- Evaluated RL algorithms (**SARSA, Q-Learning**) across configurations, identifying optimal convergence strategies.
- Published findings on hyperparameter impact to learning efficiency, guiding RL practitioners.([Article](#))

HONORS & LEADERSHIP

- **Alumni Advisor, ISA TU Delft:** Mentoring international students on navigating the Dutch higher education system and successfully transitioning into the tech industry.
- **Visionary Engineer Award (Alstom Transport):** Awarded for excellence in project planning and the optimization of system delivery schedules.
- **President, ISTE (NITK Chapter):** Led a 100+ member technical student society; managed large-scale engineering workshops and symposiums for 1,000+ participants.
- **Community Volunteer:** Organized STEM-focused educational outreach for underprivileged students in Indian government schools to bridge the digital divide.

CERTIFICATIONS

- Advanced Python Development Techniques ([LINK](#))
- Google Advanced Data Analytics Specialization ([LINK](#))
- DeepLearning.AI TensorFlow Developer Specialization ([LINK](#))
- Professional Scrum Master I ([LINK](#))

MANAGERIAL SKILLS

Project Planning

Collaboration

Customer Interaction

Agile Working

Scrum

Mentoring and Training

KNOWN LANGUAGES

English

Dutch (A2)

German (A1)

Hindi

Kannada

INTERESTS

- **Technical Writing:** Author of deep-dive articles on ML, Algorithms and Distributed Systems on **Medium** and **Substack**.
- **Sports:** Active half-marathon runner (10+ events)([Strava](#)) and Tennis
- **Creative Pursuits:** (Photography)